



Tensors for System Analysis of Converter-dominated Power Grids

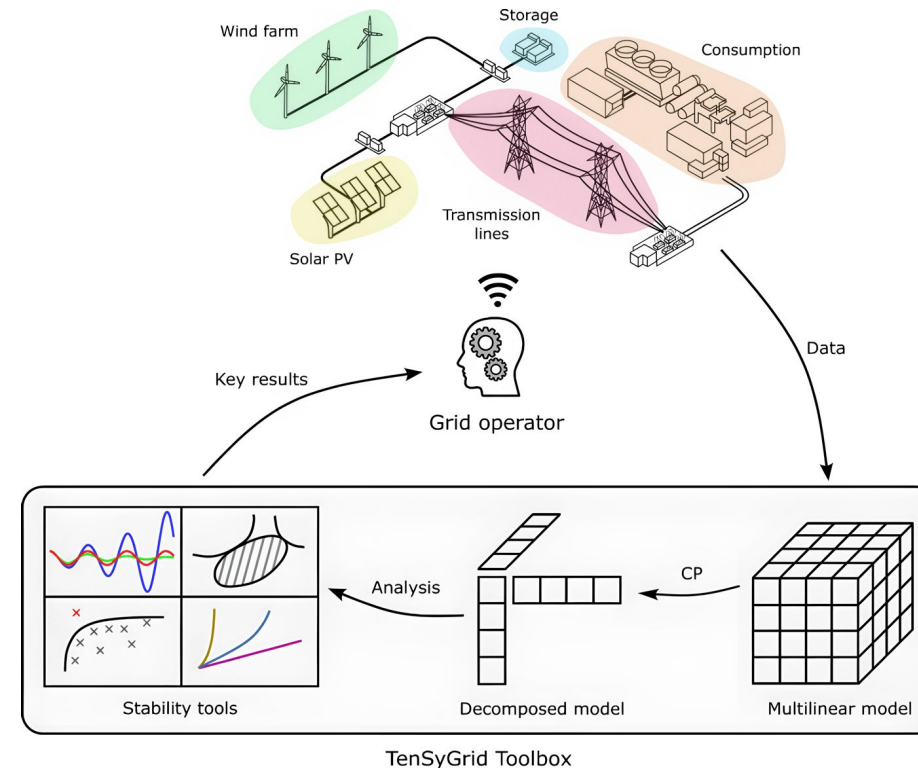
Eduardo Prieto-Araujo
Alexandre Gràcia i Calvo
Carlos Herrera Vázquez

Objectives

TenSyGrid aims to develop a toolbox with innovative stability analysis methods to facilitate the integration of 100 % renewable energy-based supply while maintaining a safe and stable operation of the power system.

Expected outputs

- Overcome *curse of dimensionality*
- Fast linearization
- Accessible toolbox integrated within open source existing tool (VeraGrid)



What is VeraGrid ?

VeraGrid [1] is a top tier power systems planning and simulation software, designed by eRoots, a spin-off founded at CITCEA-UPC.



Main features

- Open-source and 100% developed in Python
- Modular and extensible architecture (so the toolbox is just a calculation driver)
- Comprehensive analysis suite (Power Flow, OPF, Short Circuit, Dynamic & Small Signal), all in the same workflow
- Dual interface: Python API for scripting & intuitive GUI

Theoretical approach

CP decomposition

If we wanted to store all possible combinations, we would need up to 2^d of parameters, being infeasible to work with. This is known as the **curse of dimensionality**.

Instead, we can use a tensor approach called **CP decomposition**, reducing the number of parameters to just dn ($r=1, d=2$):

$$\mathcal{T} = \bigotimes_{i=1}^{n=N_v} \begin{pmatrix} 1 \\ x_i \end{pmatrix}$$

- × $\mathcal{O}(rdn), r = 1, d = 3$
- × $\mathcal{O}(rdn), r = 5, d = 3$
- × $\mathcal{O}(rdn), r = 50, d = 3$
- + $\mathcal{O}(d^n), d = 3$

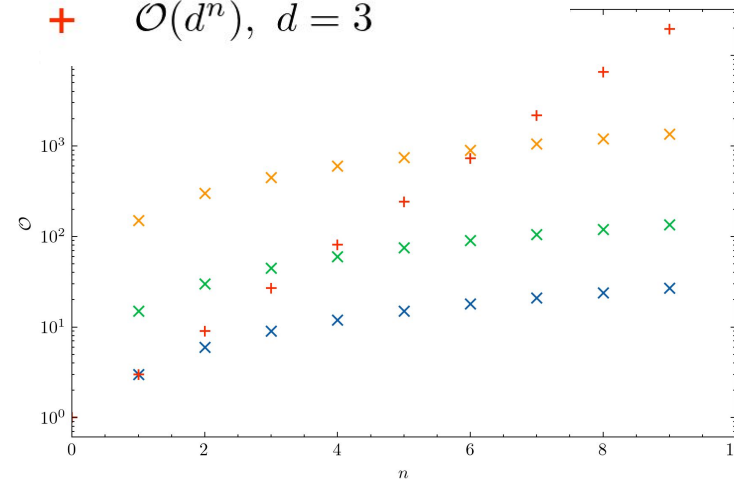


Fig. 1(a) For a fixed vector space dimension $d=\dim(V)=3$, one can see that the order of parameters to define a full tensor grows exponentially with the number of modes n while for a fixed rank r , it grows linearly.

- × $\mathcal{O}(rdn), r = 1, n = 3$
- × $\mathcal{O}(rdn), r = 5, n = 3$
- × $\mathcal{O}(rdn), r = 50, n = 3$
- + $\mathcal{O}(d^n), n = 3$

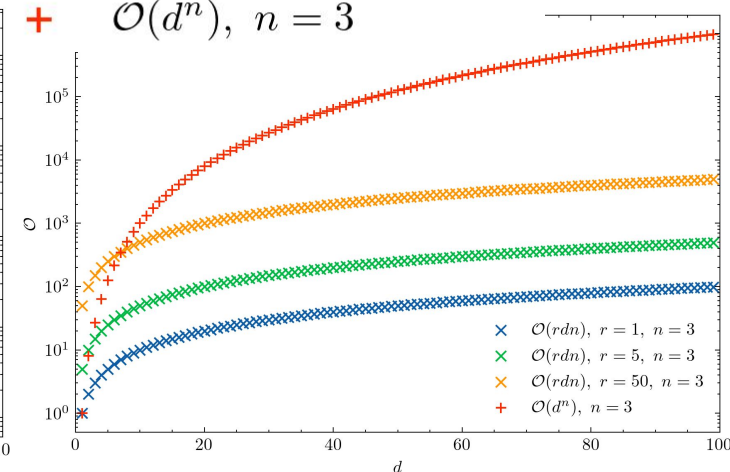


Fig. 1(b) For a fixed number of modes $n=3$, one can see that the order of parameters to define a full tensor grows cubically with the number of modes d while for a fixed rank r , it grows linearly.

Theoretical approach

Multilinear approach

The equations of the iMTI models become **multilinear** while keeping their properties [2].

- For instance,

$$x_1^2 - 1 = 0 \rightarrow$$

$$x_1 x_{1,\text{aux}} - 1 = 0$$

$$x_1 = x_{1,\text{aux}}$$

- Also valid for non-linear maps like trigonometric ones

Fast linearization

Equations are rewritten in factor matrices S and Φ .

SSA's **jacobian matrix** is built through **linear multiplications** of S and Φ thanks to multilinear approach, even though we increase the dimension of the space.

SSA is faster than usual symbolic methods [2]

Agreement with nonlinear approaches

- Factor matrices encode the equations with multilinear basis tensor
- Lie-Bläcklund transformations back the equivalence in dynamics
- Agreement when using Runge-Kutta [2]

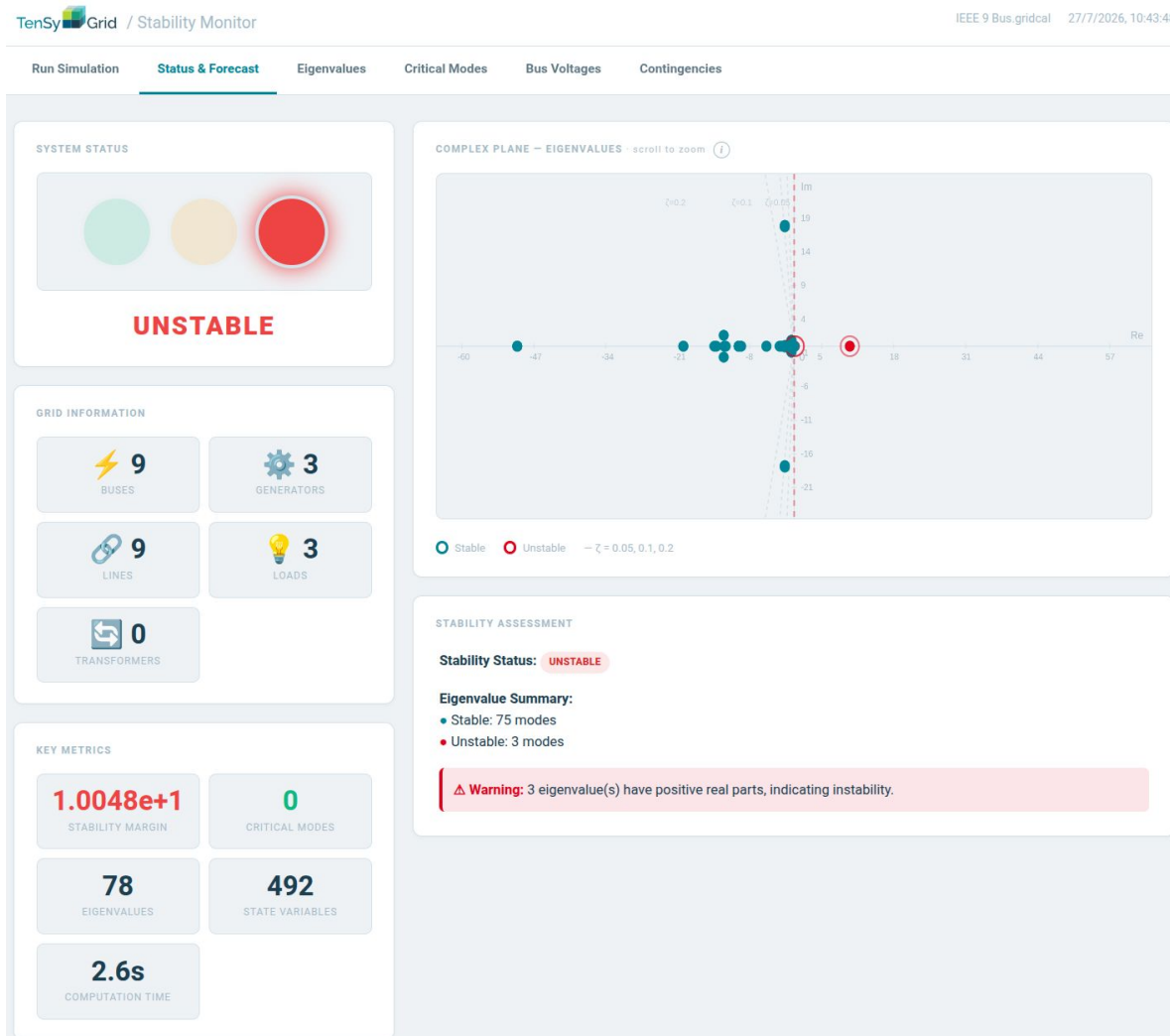
Toolbox use-cases

TenSyGrid Toolbox use case inventory [ongoing work package]

ID	Name	Status	Objective
UC-01	Run small-signal stability simulation	Implemented	Execute a stability computation on a grid model
UC-02	System status dashboard	Implemented	Obtain an at-a-glance stability classification
UC-03	Eigenvalue spectrum analysis	Implemented	Visualise and explore the eigenvalue distribution
UC-04	Critical mode diagnosis	Implemented	Identify low-damped modes and their physical origin
UC-05	Bus voltage assessment	Implemented	Verify voltage profile from the power-flow solution
UC-06	Operational decisions	Implemented	Evaluate stability impact of single-element outages
UC-07	Operating scenario comparison	Implemented	Compare stability between two grid snapshots
UC-08	Stability trend analysis	Proposed	Track stability evolution across multiple operating points
UC-09	Critical element ranking	Proposed	Rank elements by their contribution to stability degradation
UC-10	Damping controller placement guidance	Proposed	Suggest candidate locations for PSS or FACTS devices
UC-11	Renewable scenario assessment	Proposed	Evaluate stability at different renewable penetration levels
UC-12	Automated stability report generation	Ongoing	Produce a standardised stability report for stakeholders

Toolbox use-cases

UC-02. Graphical eigenvalue analysis



- General overview about small signal stability

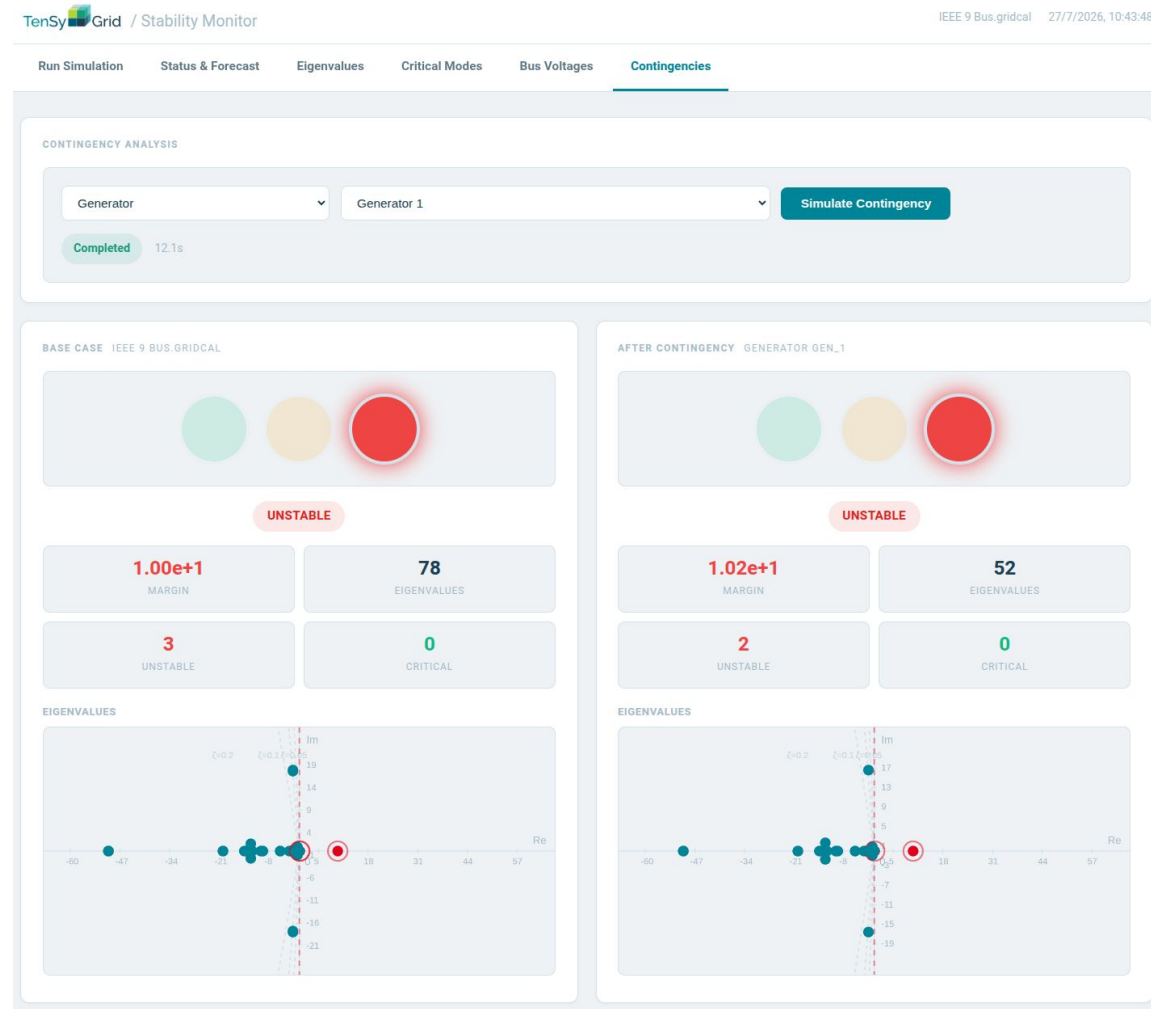
Toolbox use-cases

TenSyGrid Toolbox use case inventory [ongoing work package]

ID	Name	Status	Objective
UC-01	Run small-signal stability simulation	Implemented	Execute a stability computation on a grid model
UC-02	System status dashboard	Implemented	Obtain an at-a-glance stability classification
UC-03	Eigenvalue spectrum analysis	Implemented	Visualise and explore the eigenvalue distribution
UC-04	Critical mode diagnosis	Implemented	Identify low-damped modes and their physical origin
UC-05	Bus voltage assessment	Implemented	Verify voltage profile from the power-flow solution
UC-06	Operational decisions	Implemented	Evaluate stability impact of single-element outages
UC-07	Operating scenario comparison	Implemented	Compare stability between two grid snapshots
UC-08	Stability trend analysis	Proposed	Track stability evolution across multiple operating points
UC-09	Critical element ranking	Proposed	Rank elements by their contribution to stability degradation
UC-10	Damping controller placement guidance	Proposed	Suggest candidate locations for PSS or FACTS devices
UC-11	Renewable scenario assessment	Proposed	Evaluate stability at different renewable penetration levels
UC-12	Automated stability report generation	Ongoing	Produce a standardised stability report for stakeholders

Toolbox use-cases

UC-06-07. Operational decisions



- Manual comparison between different but similar scenarios
- Enables a fast view for an operational decision over the grid

Further work

Multilinearization for logarithms and exponentials

Since they are D-algebraic functions, an equivalent multilinear formulation should be included within this year.

- It will enable the inclusion of models for devices like electrolyzers

Results' assessment with current technologies with bigger grids

Using stable software like STAMP[3], we will check the agreement with actual expected values.

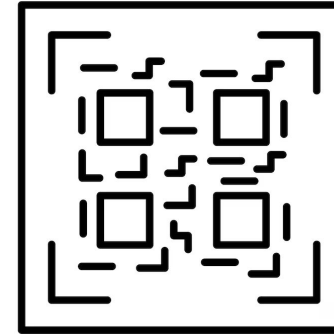
- IEEE 9 bus model and beyond
- Scaling checks (IEEE 39, IEEE 118, ACTIVSg2000)

Toolbox improvement

- Benefit from multilinear computational speed
- Identification of bottlenecks
- Adaptation of the toolbox to the market necessities

Discussion

- Ideas and comments about the toolbox are appreciated. You can access the survey here:
- You can download these slides here:





Carlos Herrera Vázquez

PhD student in Applied
Mathematics

CITCEA-UPC
Av. Diagonal 647,
Edifici G
08028 Barcelona, Spain
carlos.herrera.vazquez@upc.edu
<https://citcea.upc.edu/ca>



Eduardo Prieto-Araujo

Full professor and
principal researcher

CITCEA-UPC
Av. Diagonal 647,
Edifici G
08028 Barcelona, Spain
eduardo.prieto-araujo@upc.edu
<https://citcea.upc.edu/ca>



Alexandre Gràcia i Calvo

PhD Student in CS
and Technical support

CITCEA-UPC
Av. Diagonal 647,
Edifici G
08028 Barcelona, Spain
alexandre.gracia@upc.edu
<https://citcea.upc.edu/ca>

Thanks!

Visit us at cetpartnership.eu

Acknowledgements

This research was funded by CETPartnership, the Clean Energy Transition Partnership under the 2023 joint call for research proposals, co funded by the European Commission (GA 101 069750) and with the funding organizations detailed on <https://cetpartnership.eu/funding-agencies-and-call-modules>.

Amb el suport del programa predoctoral Joan Oró FI del Departament de Recerca i Universitats de la Generalitat de Catalunya (núm. expedient: FI 2026-1179) i el cofinançament del Fons Social Europeu Plus

Supported by:



on the basis of a decision
by the German Bundestag



MINISTERIO
DE CIENCIA, INNOVACIÓN
Y UNIVERSIDADES



AGENCIA
ESTATAL DE
INVESTIGACIÓN



Generalitat
de Catalunya



Co-funded by
the European Union

References

- [1] S. Peñate Vera et al., "SanPen/VeraGrid: 6.5.0," v6.5.0, Zenodo, ago. 2026. doi: 10.5281/zenodo.22049877.
- [2] C. Kaufmann, G. Pangalos, G. Lichtenberg, and O. Gomis-Bellmunt, "Small-Signal Stability Analysis of Power Systems by Implicit Multilinear Models," IEEE Access, vol. 14, pp. 37951–37967, 2026, doi: 10.1109/ACCESS.2026.3672335.
- [3] J. Arévalo-Soler et al., "A Matlab-based toolbox for automatic EMT modeling and small-signal stability analysis of modern power systems," arXiv:2506.22201, 2025. [Online]. Available: <https://arxiv.org/abs/2506.22201>
- [4] MATLAB, version 23.2.0 (R2023b). Natick, Massachusetts: The MathWorks Inc., 2023.
- [5] G. Lichtenberg, E. Uhlenberg, T. Warnecke, C. Cateriano Yáñez, M. Engels, L. Samaniego, L. Schnelle, and C. Kaufmann, "MTIToolbox," 2025. [Online]. Available: <https://www.mti.systems>

Support slides

Mathematical procedure

$$0 = 3\dot{x}_1 y_1 + 2z_1 x_1 - u_1 x_1 - u_1 z_1 x_1 + 2x_1$$

$$0 = \dot{x}_1 - y_1$$

$$\mathbf{v} = (\dot{x}_1 \quad x_1 \quad u_1 \quad y_1 \quad z_1)^T$$

Reconstruction via tensor contraction

$$\mathbf{s}_r(\mathbf{v}(t)) = \prod_{i=1}^{N_v} (1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir} \mathbf{v}_i(t)), \quad \mathbf{0} = \Phi \mathbf{s}(\mathbf{v}(t))$$

Structure matrix

$$\mathbf{S} = (|\langle \text{monomial vars}_j, v_i \rangle|) = \begin{pmatrix} \dot{x}_1 \text{ in } 3\dot{x}_1 y_1? & \cdots & \dot{x}_1 \text{ in } y_1? \\ x_1 \text{ in } 3\dot{x}_1 y_1? & \cdots & x_1 \text{ in } y_1? \\ \vdots & \ddots & \vdots \\ z_1 \text{ in } 3\dot{x}_1 y_1? & \cdots & z_1 \text{ in } y_1? \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Time-invariant parameter matrix

$$\Phi = \begin{pmatrix} \langle f_1, \dot{x}_1 y_1 \rangle & \langle f_1, x_1 z_1 \rangle & \cdots & \langle f_1, y_1 \rangle \\ \langle f_2, \dot{x}_1 y_1 \rangle & \langle f_2, x_1 z_1 \rangle & \cdots & \langle f_2, y_1 \rangle \end{pmatrix} = \begin{pmatrix} 3 & 2 & -1 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

Jacobian construction

Reconstruction (inefficient for loops)

$$\left. \frac{\partial s_r(\mathbf{v})}{\partial v_i} \right|_{\mathbf{v}=\bar{\mathbf{v}}} = \mathbf{S}_{ir} \prod_{\substack{j=1 \\ j \neq i}}^{N_v} (1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr} \bar{v}_j)$$

Jacobian

$$\mathbf{J} = \Phi(\mathbf{D} \boxtimes \mathbf{s}(\bar{\mathbf{v}}))^T \in \mathbb{R}^{N_\phi \times N_v}$$

$$\mathbf{D}_{ir} = \begin{cases} \frac{\mathbf{S}_{ir}}{1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir} \bar{v}_i} & \text{iff } s_r(\bar{\mathbf{v}}) \neq 0, \forall r = 1, \dots, R, \\ \mathbf{S}_{ir} & \text{iff } (1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir} \bar{v}_i) = 0, \\ 0 & \text{else,} \end{cases}$$

$$s_r(\bar{\mathbf{v}}) = \begin{cases} \prod_{j=1}^{N_v} (1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr} \bar{v}_j) & \text{iff } s_r(\bar{\mathbf{v}}) \neq 0, \forall r = 1, \dots, R, \\ \prod_{\substack{j=1 \\ j \neq i}}^{N_v} (1 - |\mathbf{S}_{jr}| + \mathbf{S}_{jr} \bar{v}_j) & \text{iff } (1 - |\mathbf{S}_{ir}| + \mathbf{S}_{ir} \bar{v}_i) = 0, \\ 0 & \text{else.} \end{cases}$$

Lie-Bläcklund isomorphism

$$\begin{aligned}
 x &\leftrightarrow x \\
 \frac{dx}{dt} &\leftrightarrow dxdt \\
 \cos(x) &\leftrightarrow u_{\cos} \\
 \sin(x) &\leftrightarrow u_{\sin} \\
 \frac{d \cos(x)}{dt} &\leftrightarrow -u_{\sin} dxdt \\
 \frac{d \sin(x)}{dt} &\leftrightarrow u_{\cos} dxdt
 \end{aligned}$$

+

$$u_{\cos}^2 + u_{\sin}^2 - 1 = 0$$

This restriction is imposed to guarantee the diffeomorphism.

This equation should also to be multilinearized!

(Non/multi)-linear comparison

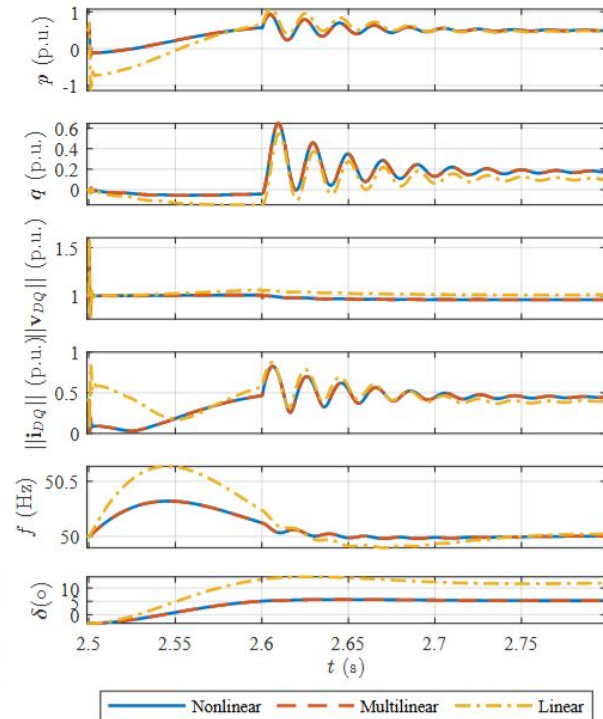
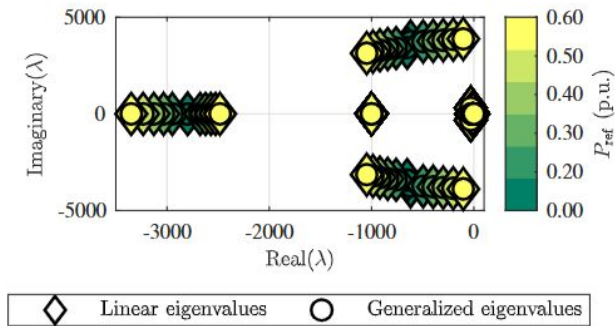
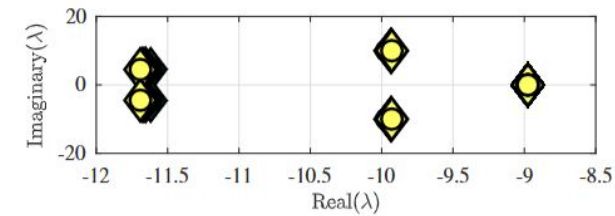


FIGURE 10: Response of the grid-forming converter to a 50 % load reduction at 2.5 s and 9 % voltage drop at 2.6 s



(a)



(b) Zoom plot of (a)

FIGURE 13: Comparison of the generalized eigenvalues of the linearized iMTI model with the eigenvalues of the linearized NTI model

Speed up comparison [2]

Model properties					Control System Toolbox by Matlab [4]		MTI Toolbox [5]	
n Number of state variables	m Number of input variables	$p+q$ Number of output + algebraic variables	R Number of monomial	nnz (%)	lin (ms)	eig (ms)	lin (ms)	eig (ms)
a) 16	6	17	87	3.09	309.7	0.123	0.0133	0.1420
b) 15	6	17	95	3.57	450.34	0.1091	0.0144	0.1221
c) 26	8	32	165	2.03	471.02	0.257	0.0474	0.4776

[2] C. Kaufmann et al., IEEE Access, vol. 14, pp. 37951–37967, 2026.

[4] MATLAB, version 23.2.0 (R2023b). Natick, Massachusetts: The MathWorks Inc., 2023.

[5] G. Lichtenberg et al., “MTIToolbox,” 2025. [Online]. Available: <https://www.mti.systems>